E-Commerce Sales and Customer Behaviour Analysis

Exploratory Data Analysis Project

1. Introduction

This project explores the UCI Online Retail dataset to understand transaction patterns, product performance, customer spending and sales trends.

2. Dataset Overview

The original dataset contains 541,909 records and eight columns, including invoice number, product code, description, quantity, invoice date, unit price, customer ID and country.

3. Data Preparation

Exact duplicate rows were removed, along with rows without product descriptions. Customer-level analysis excludes records without customer IDs. The completed-sales dataset excludes cancelled invoices, non-positive quantities and non-positive unit prices.

After these steps, 392,692 transaction lines remained in the completed-sales dataset.

4. Analysis and Findings

Total revenue in the completed-sales dataset was £8,887,208.89. The analysis contained 18,532 distinct completed invoices and 4,338 customers. Average invoice value was £479.56.

The United Kingdom generated the highest revenue. Among countries outside the UK, the Netherlands generated the highest revenue. November 2011 was the highest-revenue month among the months shown in the analysis.

PAPER CRAFT, LITTLE BIRDIE was the highest-revenue product in the analysis. Thursday recorded the highest total revenue among weekdays. The correlation between quantity and line revenue was approximately 0.91.

5. Interpretation

Revenue varies across products, countries and months. The strong positive relationship between quantity and line revenue is consistent with larger quantities contributing to higher transaction-line revenue. Product prices and the distribution of purchases also matter.

The observed cancellation rate was calculated from the customer-linked dataset and should be interpreted with that denominator in mind.

6. Business Recommendations

1. Monitor high-revenue products and maintain appropriate stock levels.

2. Compare country-level performance to identify opportunities outside the UK.

3. Investigate seasonal revenue changes before making inventory decisions.

4. Review cancellation patterns to identify possible operational issues.

5. Use customer-level purchasing behaviour to develop appropriate retention strategies.

7. Limitations

The analysis describes historical transactions and does not establish causal relationships. Revenue reflects the cleaning decisions described in the notebook. Missing customer IDs limit customer-level analysis, and the dataset may not represent current retail behaviour.

8. Conclusion

The analysis identifies important differences in revenue across products, countries and time periods. These findings can support further investigation of inventory planning, customer retention and international sales opportunities.

Dataset source: UCI Machine Learning Repository, Online Retail. Report values are based on the results provided during the analysis workflow; verify them against the saved notebook before final submission.